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Toward the Development of an RRQL System—Part I: Swirl Pattern Effect on Exhaust Emissions and Chemiluminescence Distribution for NH_3–Air Premixed Swirl Flames

Ammonia (NH_3) is a carbon–free energy source and hydrogen carrier, but its fuel–bound nitrogen can lead to significant nitrogen oxides (NO_x) emissions. Staged combustion strategies, such as rich–quench–lean, can achieve low NO_x emissions. However, improper design may result in high NO_x levels when operating with a rich head end without a sufficient post–flame relaxation time. Previous work has shown that a rich–relaxation–quench–lean (RRQL) configuration can minimize NO_x emissions in the rich head end (Cole et al., 2024, "Rich Ammonia Flame Shapes and NO Relaxation: Facility Development and Characterization," ASME Paper No. GT2024–122369). This study focuses on how different swirl geometries affect exhaust emissions and flame morphology in rich, premixed NH_3–air flames, for the design of a rich relaxation head end of an RRQL combustor. Experiments were conducted using a modular swirl burner and measuring NO_x, N_2O, and NH_3 emissions, and recording natural flame luminosity and NH_2 and OH* chemiluminescence images. Swirler design affects emissions by influencing flame length (with shorter flames allowing for more postflame relaxation time) and flame–wall interactions (influencing NH_3 and N_2O emissions). A compact flame with minimal wall interactions allows for increased NO_x relaxation, while minimal wall interactions prevent heat losses and instabilities, which minimize NH_3 and N_2O emissions at a richer equivalence ratio, potentially enhancing H_2 production during the relaxation phase without increasing overall primary stage NO_x emissions, thereby improving system efficiency. Detailed spatial evolution of NO_x, N_2O, and NH_3 emissions further supports the RRQL as a promising combustor design for NH_3 combustion, if the rich head end is designed properly.* [DOI: 10.1115/1.4069796]

1 Introduction

Ammonia (NH_3) is gaining interest as a potential noncarbon containing energy source to contribute to decarbonizing the energy sector [1–3]. Its well–established production, storage, and distribution infrastructure—largely driven by applications in fertilization and refrigeration—positions it as a feasible alternative to fossil fuels. However, NH_3 combustion presents significant challenges, including flame stability and emissions control. The slow flame speeds, low reactivity, and narrow flammability limits lead to stability and operability concerns. From an emissions perspective, the additional fuel–bound nitrogen oxides (NO_x) penalty, NH_3 slip, and nitrous oxide (N_2O) emissions lead to environmental impact concerns. Different combustion schemes, including fuel blends, high swirl stabilization, and staged combustion strategies, among others, have been proposed by researchers as potential avenues to mitigate the reactivity and emissions challenges with NH_3 [4–6].

Chemiluminescence studies for NH_3–air flames can map radical intensities with flame shapes and may potentially be correlated with emissions trends in a reaction zone in a non–invasive manner. For example, chemiluminescence studies have shown that N_2O and NO emissions have been correlated with NH_2 radicals [7]. NH_2 is a key species in NO_x formation and reduction through HNO reactions, as well as NH_3 oxidation through NH_i pathways [8,9]. Additionally, OH* is known to accelerate NH_3 decomposition, and NO_x

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formation in the flame zone is sensitive to OH* production [10], and Zhu et al. found that OH* intensity correlates with peak NO fractions for NH_3–H_2 blends [11].

Researchers at Cardiff University have extensively examined NH_3–H_2 and NH_3–CH_4 blends at ratios that yield comparable flame speeds to typical hydrocarbons. Pugh et al. studied single–stage swirling NH_3–H_2–air flames and found operability limits narrower than other fuels but with relatively good flame stability [5,12]. From an emissions perspective, they reported that rich equivalence ratios produced minimal NO_x levels and high NH_3 slip, and NH_2^* chemiluminescence was observed to peak at similar conditions. Additionally, N_2O emissions rose in extreme lean cases, whereas NH* and OH* were found to peak closer to stoichiometric lean cases.

Hayakawa et al. also extensively studied swirling NH_3–air flames [13]. The increased swirl intensity enhanced reactant mixing but led to higher NO_x emissions, which were attributed to higher flame temperatures. NO_x emissions were minimized when NH* would peak under rich conditions. In contrast, OH* exhibited a maximum, and NO_x levels were elevated under lean conditions.

Wang et al. studied how residence time and swirl intensity affect NO_x emissions in premixed NH_3–CH_4–air swirling flames using a bespoke single–stage swirl burner at elevated pressures [14]. Increasing residence time, either by extending the combustion chamber or enhancing swirl, was found to advance chemical reactions and reduce NO_2 and N_2O emissions, with swirl showing greater efficiency in lowering NO_x than increasing combustor length. However, increased swirl can slightly decrease combustion efficiency, as evidenced by increased CO under certain conditions. These findings provide insight into optimizing flame configurations to minimize NO_x emissions without compromising performance and disagree with findings from Hayakawa et al. [13].

Cole et al. recently introduced the rich–relaxation–quench–lean (RRQL) strategy to minimize NO_x emissions [15]. The relaxation stage takes advantage of relatively low equilibrium NO_x emissions produced from rich NH_3–air flames and utilizes long residence times and elevated pressures to kinetically relax NO_x emissions toward equilibrium values [4]. The RRQL experimental facility demonstrated NO_x relaxation in the primary stage, which consisted of a swirling NH_3–air flame in a modular swirl burner. A preliminary study of the effect of swirl number (S_g) on flame stability and shape is presented; however, the effect of swirl geometry on emissions was left for future studies. Several new 3D–printed modular swirlers were studied, examining the effects of S_g, number of vanes, and vane shape on flame morphology, NH_2^* and OH* radical distribution, and averaged emissions at a single residence time. Finally, detailed axial and radially resolved emissions are compared between the standard swirler design [15] and the best performing swirler geometry.

2 Materials and Methods

2.1 Burner and Conditions. Experiments were conducted with a lab–scale modular burner, shown in Fig. 1. The burner consists of a modular swirler, and the swirler center body houses a hydrogen torch and spark ignitor which is used only to ignite the NH_3–air mixture. This burner simulates the rich and relaxation sections of an RRQL NH_3 combustor, and the reader can refer to Ref. [15] for further details. The burner is built to accommodate additional upgrades for pressurized lean stage combustion after the rich burn and relaxation stages. It has also been designed to accommodate a modular and interchangeable swirler, which is 3D–printed stainless steel with diverse geometric swirl numbers (S_g) and swirler geometries. They are each either axial curved or straight

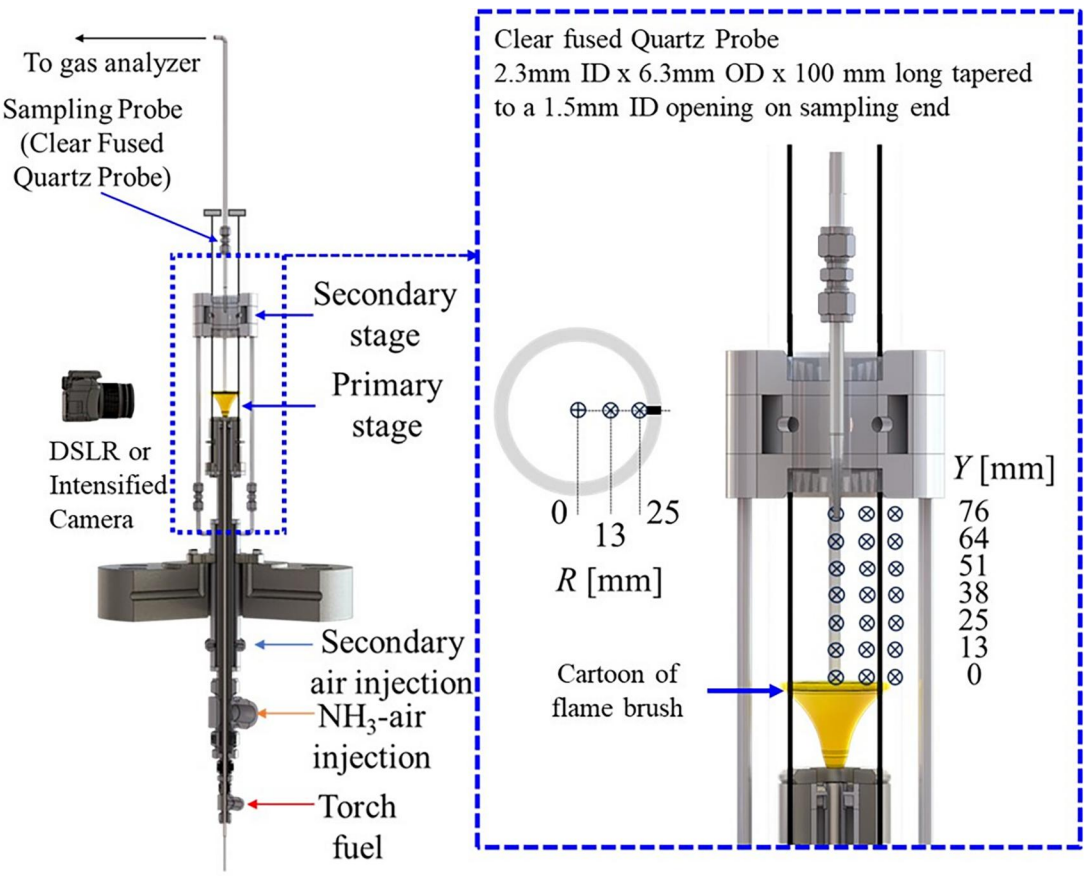

Fig. 1 Burner schematic with diagnostics

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Table 1 Definition and parameter of several swirlers tested

Tag	S_g	No. of vanes	Vane type	Vane angle (deg)	Effective area (m^2)
S_g*0.7–S16*	0.7	16	Straight	43.25	6.7×10^{-4}
S_g*1.1–C8*	1.1	8	Curved	45.18	8.8×10^{-4}
S_g*1.1–C12*	1.1	12	Curved	35.90	7.8×10^{-4}
S_g*1.1–S12*	1.1	12	Straight	55.95	7.8×10^{-4}
S_g*1.1–S16*	1.1	16	Straight	55.95	6.7×10^{-4}

vane swirlers, and S_g is calculated using Eq. (1) [16] for curved vanes and Eq. (2) [16] for flat vanes.

$$S_g = \frac{1}{1-\varphi} * \frac{1}{2} * \frac{1-\left(\frac{D_{\text{hub}}}{D_{\text{sw}}}\right)^4}{1-\left(\frac{D_{\text{hub}}}{D_{\text{sw}}}\right)^2} \tan\theta;\ \varphi = \frac{N_{\text{vanes}} * t_{\text{vane}}}{2 * \pi * R_{\text{hub}} * \cos\theta} \tag{1}$$

$$S_g = \frac{2}{3} * \frac{1-\left(\frac{D_{\text{hub}}}{D_{\text{sw}}}\right)^3}{1-\left(\frac{D_{\text{hub}}}{D_{\text{sw}}}\right)^2} \tan\theta \tag{2}$$

Table 1 shows that the five different swirlers were tested with various S_g and different numbers of vanes (8, 12, 16). Due to the change in effective area, the thermal power has been slightly adjusted between 0.75 and 0.85 kW to keep a constant nozzle velocity (U_0) of 0.90 m/s. The nozzle velocity is calculated by considering mass continuity of the air–fuel mixture across the effective area of the swirler, and the swirler effective area is calculated to account for swirler vane blockage. Swirlers are named for their S_g (0.7 or 1.1), geometry of the vanes (straight or curved), and number of vanes, e.g., S_g*0.7–S16* corresponds to a swirler with $S_g = 0.7$ and 16 straight vanes.

The combustor liner consists of a 33 cm long quartz tube with a 5.02 cm inner diameter that fits concentrically onto the burner and seals with an O–ring. The length from the dump plane to the top of the quartz is 26.5 cm. There is a blockage plate on the top of the quartz to prevent backflow or interactions with outside air. The long quartz tube is meant to allow NO_x emissions to relax toward equilibrium with residence times around 100 ms post–flame for a 5 cm flame, depending on the nozzle velocity and expansion ratio. As shown in Ref. [4], these values should enable the capture of the first order of magnitude NO relaxation at 1 bar and nearly the full approach to equilibrium at 10 bar.

Equivalence ratios (ϕ) and the nozzle velocity are calculated in real–time using the mass flow ratios for the fuel and air circuits. The mass flow rates are metered using manual and electronic pressure regulators to control upstream pressures through critical orifices. An upstream pressure transducer and thermocouple in combination with a downstream pressure transducer allow the operator to ensure sonic flow for effective metering. An additional thermocouple is located upstream of the swirler to track the reactant temperature. The air and fuel flow control setup described leads to an accuracy better than 1% for the ϕ. The nozzle velocity is defined using the reactant density, effective nozzle area, and reactant mass flow. The Reynolds number at the nozzle exit varies between 10,000 and 18,000, and it is most sensitive to nozzle velocity and varies only marginally with equivalence ratio due to similarities in kinematic viscosity for air and NH_3.

Table 2 Experimental matrix

Parameter	Value
Equivalence ratio (ϕ)	1.05/1.10/1.13/1.15/1.18
Nozzle velocity (U_0)	0.90 m/s
Geometric swirl (S_g)	0.7/1.1
Inlet temperature	298 K
Inlet pressure	0.101 MPa
Ammonia mole fraction (X_{NH3})	1.0

2.2 Exhaust Emissions. Emissions data are collected using a California Analytical Instruments 700 LX Series non–dispersive infrared (NDIR) O_2 Analyzer for O_2 and 700 Fourier–transform infrared spectroscopy (FTIR) multigas analyzer with NO, NH_3, and N_2O measurement capabilities (Orange, CA). The emissions sample line between the probe and analyzer is heated to 150 °C. Exhaust emissions were normalized by the standard of 15% dry O_2 [17]. A single–point quartz emissions probe was used for all emissions measurements. For the average emissions comparison, as shown in Sec. 3.3, the three radial points are area averaged. For a more detailed investigation of NO_x relaxation, the emissions are resolved radially at three positions: centerline, midway between the wall and centerline, and a near–wall point, for seven axial locations spanning from the flame brush, determined by the NH_2^* chemiluminescence isocontour, to 7.6 cm downstream from the flame brush. As shown in Fig. 1, a clear fused quartz sampling probe with a long tapered end (1.5 mm diameter) was implemented to withdraw gas samples both at and downstream from the flame brush, and the reader is referred to

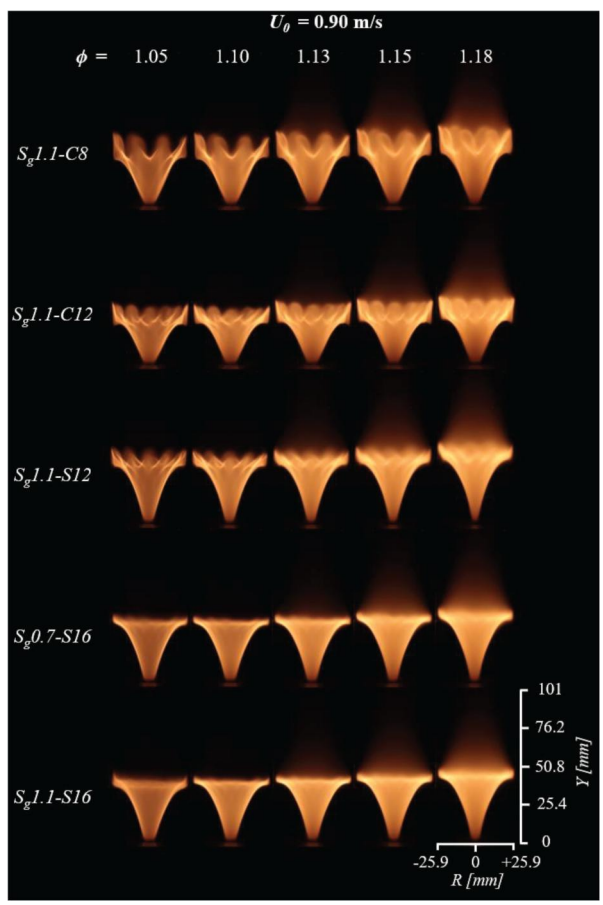

Fig. 2 DSLR flame images for test matrix of modular swirler geometries for various ϕ

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Ref. [18] for additional information about the sampling probe design. A condenser is located downstream from the probe and upstream from the emissions bench, which hosts the FTIR and NDIR modules. The condenser dries the sample to protect the analyzers and prevent any interference with the FTIR spectra. The authors acknowledge that NH_3 can be absorbed by water, and sensitivity studies were conducted to show that less than 100 ppm of NH_3 may be absorbed by the water in the condenser. In addition, the sample is kept at 150 °C to avoid water condensation inside the line and equipment.

2.3 Flame Imaging. Time–averaged broadband flame images were recorded with a digital single-lens reflex (DSLR) camera with an exposure time of 1/4 s, an aperture of *f*/16, and an ISO number of 800. White balance was kept constant at 4000 K throughout the experimental campaign. Chemiluminescence images were recorded with a high–speed, intensified camera (FASTCAM (Tokyo, Japan) SA–X2/HiCATT intensifier) fitted with a UV lens (CERCO® (Concord, CA), 2178, $f = 150$ mm, *f*/2.8). A 20 nm bandpass filter centered at 308 nm (Invisible Vision) was used for OH*, and a 10 nm bandpass filter centered at 633 nm (Andover Corp., Salem, NH, 633FS10-50) was used for NH_2^*. It is acknowledged that several species may emit in this range, such as H_2O^*, NO_2^*, and NH_2^* [19,20]. The PHOTRON FASTCAM software is used to record 300 frames for each data condition at 500 fps, which were then background subtracted, temporally averaged, and inverse Abel–transformed during post-processing to map flame shapes across the swirlers under the varying swirlers and equivalence ratios listed in Table 2.

3 Results and Discussion

3.1 Flame Appearance. Figure 2 shows the time–averaged broadband flame images as a function of the ϕ for all the swirlers tested. In general, the flames present the typical V–shaped morphology of a swirling flame, and the flame length increases as a function of the ϕ. The $S_g1.1$–$C8$ swirler (first row) produces a compact flame with lobe–like structures when the flow interacts with the combustor liner walls. Generally, compact flames are desirable due to their increased NO_x relaxation zones for a fixed geometry. However, in a preliminary study, flame instabilities and opportunities for improved emissions production [15] were found for this specific swirler. These lobes decrease by increasing the number of vanes ($S_g1.1$–$C12$), but they are still present. Using a swirler with straight (flat) vanes sharpens the lobe—like structures but produces a slightly more elongated flame ($S_g1.1$–$S12$). Increasing the number of vanes to 16 ($S_g0.7$–$S16$) produces a compact flame, and minor differences can be spotted for the two swirl numbers $S_g0.7$–$S16$ and

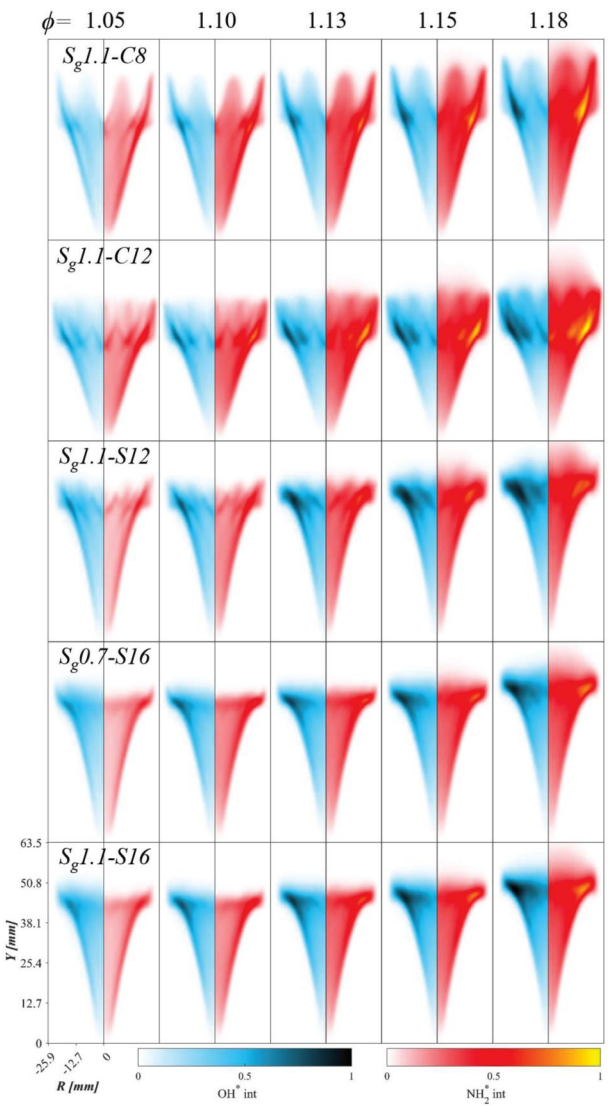

Fig. 3 Mean normalized OH* (left) and NH_2^* (right) as a function of equivalence ratio (left to right) and swirler (top to bottom)

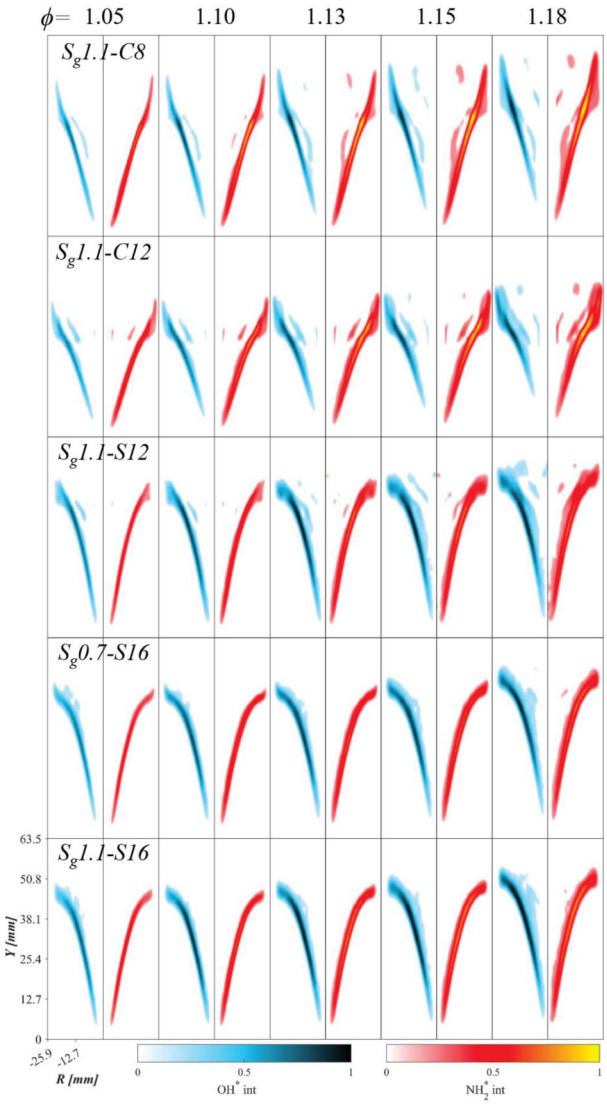

Fig. 4 Inverse Abel–transformed OH* (left) and NH_2^* (right) as a function of equivalence ratio (left to right) and swirler (top to bottom)

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$S_g1.1$–$S16$, which produce flames with minor flame to wall interaction. Increased flame to wall interaction was found to increase flame sensitivity to quartz–burner alignment and triggered an intermittent precessing outer recirculation zone kernel which leads to increased NH_3 emissions.

3.2 Flame Chemiluminescence. Figure 3 presents the mean normalized OH* (left panel) and NH_2^* (right panel) as a function of ϕ for all the swirlers tested. OH* is a marker for high temperature reaction zones, while NH_2^* is associated with NH_3 decomposition and intermediate reactions. All the flames are inner shear layer stabilized in the inner recirculation zone; however, increasing ϕ changes the spatial distribution and intensity of OH* and NH_2^* [21]. At lean conditions, both radical concentrations increase across the flame surface and concentrations in regions near the edges, where the flame would interact with the quartz wall. Changing swirler geometry also captures radical concentrations throughout the visible flame geometry.

Figure 4 presents the normalized inverse Abel–transformed OH* (left panel) and NH_2^* (right panel) as a function of ϕ for all the swirlers tested. The Abel–transformed images show increasing radical concentration with ϕ across the different swirler geometries. Notably, geometries $S_g1.1$–$C8$, $S_g1.1$–$C12$, and $S_g1.1$–$S12$ show a double walled structure which is an artifact of the lobes of the swirler.

Figure 5 presents isocontours for inverse Abel–transformed OH* (black lines) and NH_2^* (gray lines) as a function of ϕ for all the swirlers tested. The solid lines represent isocontours of 40% maximum intensity, and the dashed lines represent isocontours of 90% intensity. Overall, the NH_2^* radicals are primarily observed on the outer side of the flame front, beginning at the swirler outlet and extending downstream. Looking at the 40% intensity isocontour, as ϕ increases, the NH_2^* profile becomes more pronounced and lifts slightly, indicating a delayed stabilization of the flame in richer mixtures. Additionally, the length of the NH_2^* zone grows with increasing ϕ, signifying enhanced radical production in the preheat zone. In contrast, OH* radicals are concentrated on the inner side of the flame front and are typically lifted, reflecting their association with higher temperature regions closer to the flame core. Furthermore, when analyzing the 90% isocontour line, the relative locations of NH_2^* and OH* are consistent, where OH* is typically lifted on the inner edge of the flame.

Generally, with increasing ϕ, the length of the radical isocontour decreases, which is particularly apparent for the swirl geometries with a visible lobe structure ($S_g1.1$–$C8$, $S_g1.1$–$C12$, and $S_g1.1$–$S12$). Furthermore, these swirl geometries with lobe structure present in the flame brush have a smaller area in the 90% isocontour for both radical distributions as opposed to the swirlers, which produce a more traditional inner recirculation zone flame shape ($S_g1.1$–$S16$, $S_g0.7$–$S16$). The key takeaway in terms of swirler geometry performance from the chemiluminescence images is the interaction of the lobe structures with the quartz liner. Regions of high OH* intensity, such as the 90% OH* isocontour, are located closer to the quartz liner, which can cause dramatic effects on NO_x relaxation, NH_3 cracking, and N_2O emissions produced due to cold liner walls.

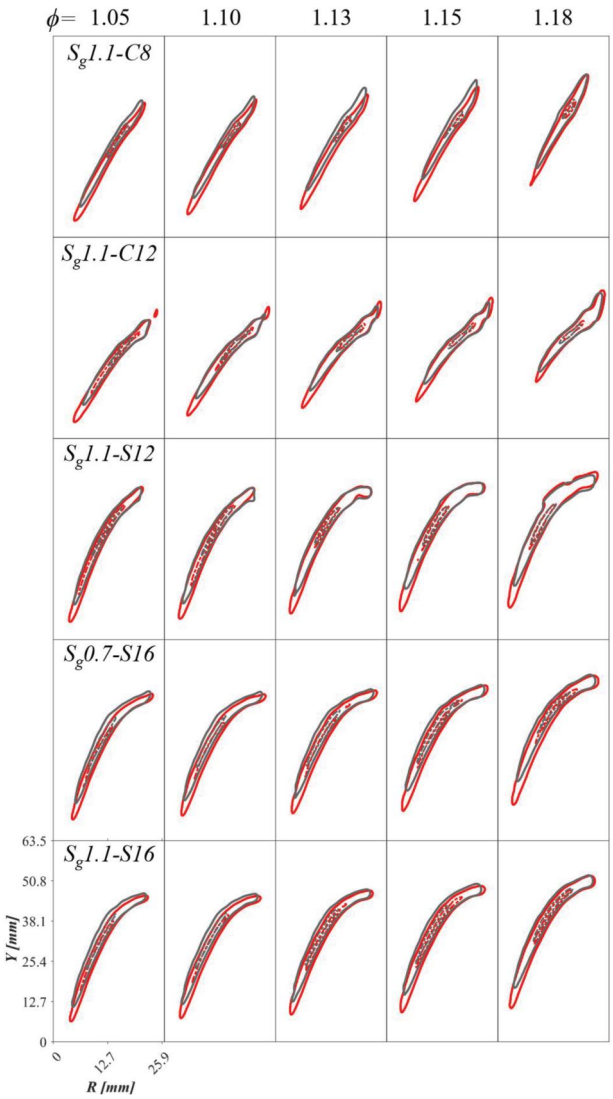

Fig. 5 Isocontours for 40% (solid lines) and 90% (dashed lines) intensity of inverse Abel–transformed OH* (black dashed line) and NH_2^* (gray line) images as a function of equivalence ratio and swirler

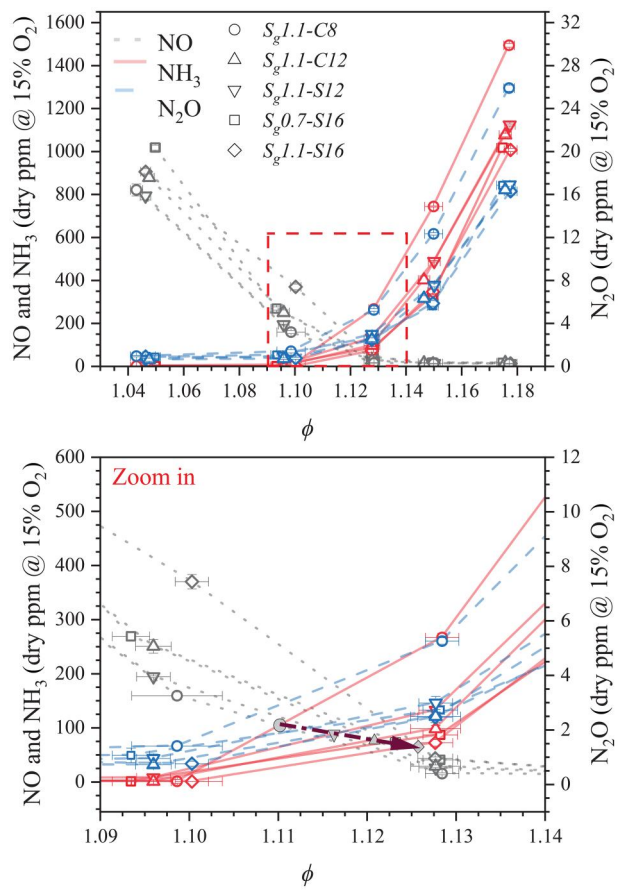

Fig. 6 NO_x, N_2O, and NH_3 concentration in the flue gases as a function of the equivalence ratio. Zoom in to show the sweet spot for low NO_x and NH_3 concentration.

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3.3 Exhaust Emissions. Figure 6 shows the concentration of NO_x, NH_3, and N_2O at 7.6 cm downstream from the flame, or ~80 ms from the flame brush, as a function of ϕ for all the swirlers tested. It is observed that in general NO_x decreases while N_2O and NH_3 increase as a function of ϕ, trends that are well known in literature [10]. To better observe the impact of the swirler geometry on the optimal ϕ (minimum NO_x–NH_3 tradeoff), a zoomed-in plot is also shown in Fig. 6 (bottom panel). It can be seen that $S_g1.1$–*C8* has its sweet spot at a leaner ϕ (see circle solid dot) at approximately 1.11. This optimal spot shifts toward richer ϕ (see red arrow) by increasing the number of vanes. The vane shape, curved versus straight, causes a slight shift in the optimal ϕ toward the rich end, which can be compared with the $S_g1.1$–*C12* and $S_g1.1$–*S12* geometries. Furthermore, increasing S_g also causes a slight shift in the optimal ϕ toward the rich end, which can be compared with the $S_g1.1$–*S16* and $S_g0.7$–*S16* geometries. However, this shift is within the average error in ϕ. Although not demonstrated within this test matrix, the S_g will affect flame stability and decrease the stable operability limits. Increasing the number of vanes to 16 in the $S_g1.1$–*S16* produced the lowest NO_x emissions at the richest sweet spot of 1.13 (see diamond solid dot) which could produce more H_2 for the second stage.

Generally, with improved flame shape and decreased liner interaction caused by slight modifications in swirler geometry such as the number of vanes and the vane geometry, the optimal ϕ shifts to richer equivalence ratios. The flame shape and peak OH* concentrations show that with increased heat losses to liner walls, at the same ϕ NH_3 and N_2O emissions double. NO_x emissions see a slight improvement due to the increased NH_3 slip, meaning that all the fuel–bound N_2 is not being released in the primary stage.

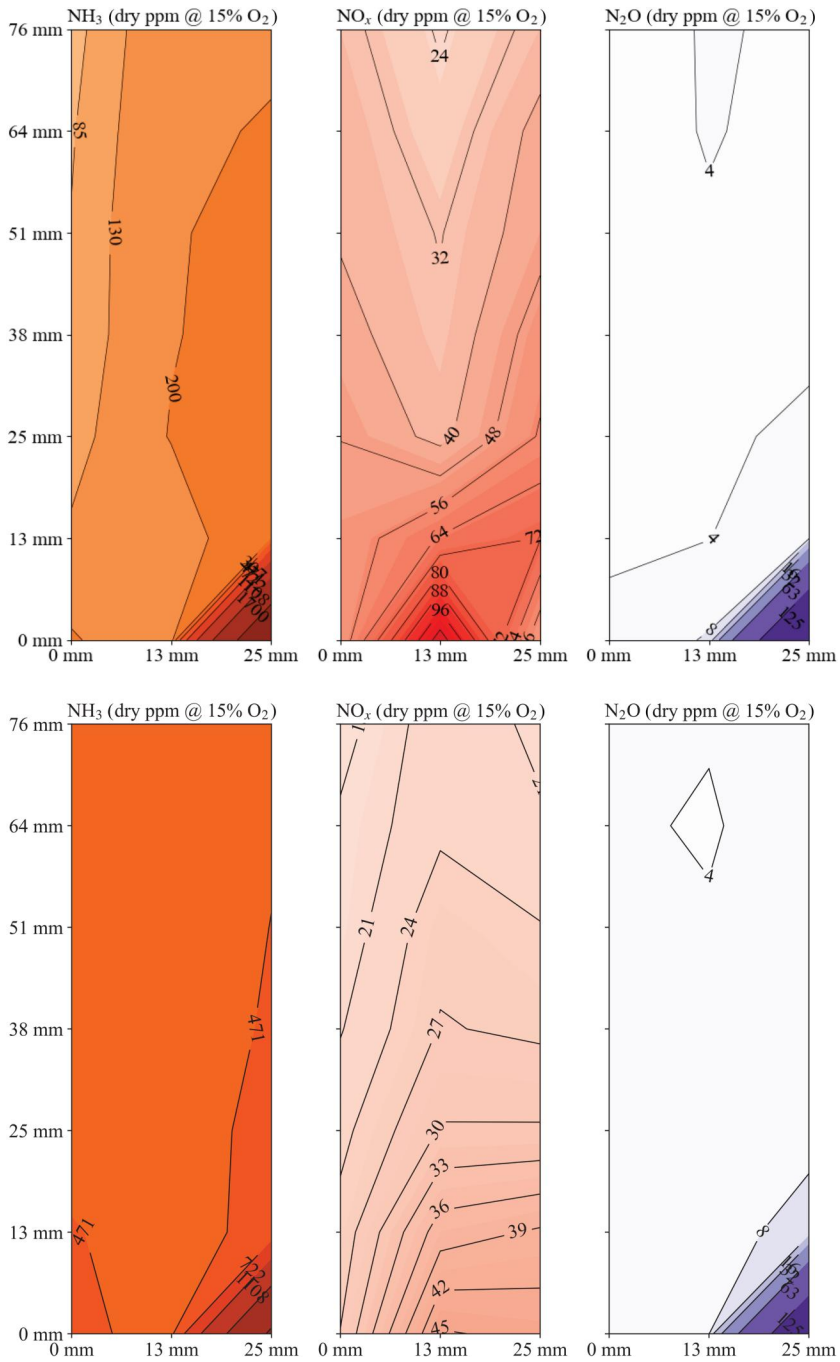

Fig. 7 NO_x, N_2O, and NH_3 emissions at 15% dry O_2 corrected for $S_g1.1$–*S16* (top) and $S_g1.1$–*C8* (bottom) at $\phi = 1.13$

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However, in a staged combustion architecture, such as RRQL, the NO_x emissions in the second stage would be expected to spike significantly due to the increased NH_3 slip, and NO_x emissions would not have the opportunity to relax toward equilibrium values.

3.4 NO_x Relaxation. NO_x, NH_3, and N_2O emissions were recorded for the 30 spatially resolved measurements as described in Fig. 1 for two different ϕ (1.13 and 1.15) for the two swirlers, $S_g1.1$–*C8* and $S_g1.1$–*S16*, which had the leanest and richest optimal ϕ, respectively. $S_g1.1$–*C8* has an optimal primary equivalence ratio of 1.11, and $S_g1.1$–*S16* has an optimal primary equivalence ratio of 1.13. The ϕ sweeps were predetermined based on previous studies that have used insulation to decrease heat losses in the quartz and are therefore slightly richer than the optimal swirler ϕ [15]. Figure 7 represents the emissions plot for $S_g1.1$–*S16* and $S_g1.1$–*C8* at ϕ of 1.13, and Fig. 8 represents both swirlers at ϕ of 1.15. For both swirler geometries, elevated emissions are located at the quartz wall at the visible flame brush, or the most downstream observable region of the flame, where reactions occur. The spike in all emissions species confirms the axial location of the flame brush from the flame imaging.

Qualitatively, both swirlers demonstrate similar trends in terms of NO_x relaxation as a percentage of maximum NO_x measurement at the flame brush. Both swirlers generate compact flames with long NO_x relaxation zones within the test matrix. However, the $S_g1.1$–*S16* geometry demonstrates lower concentrations of NH_3 slip post flame brush, indicating lower heat losses and higher

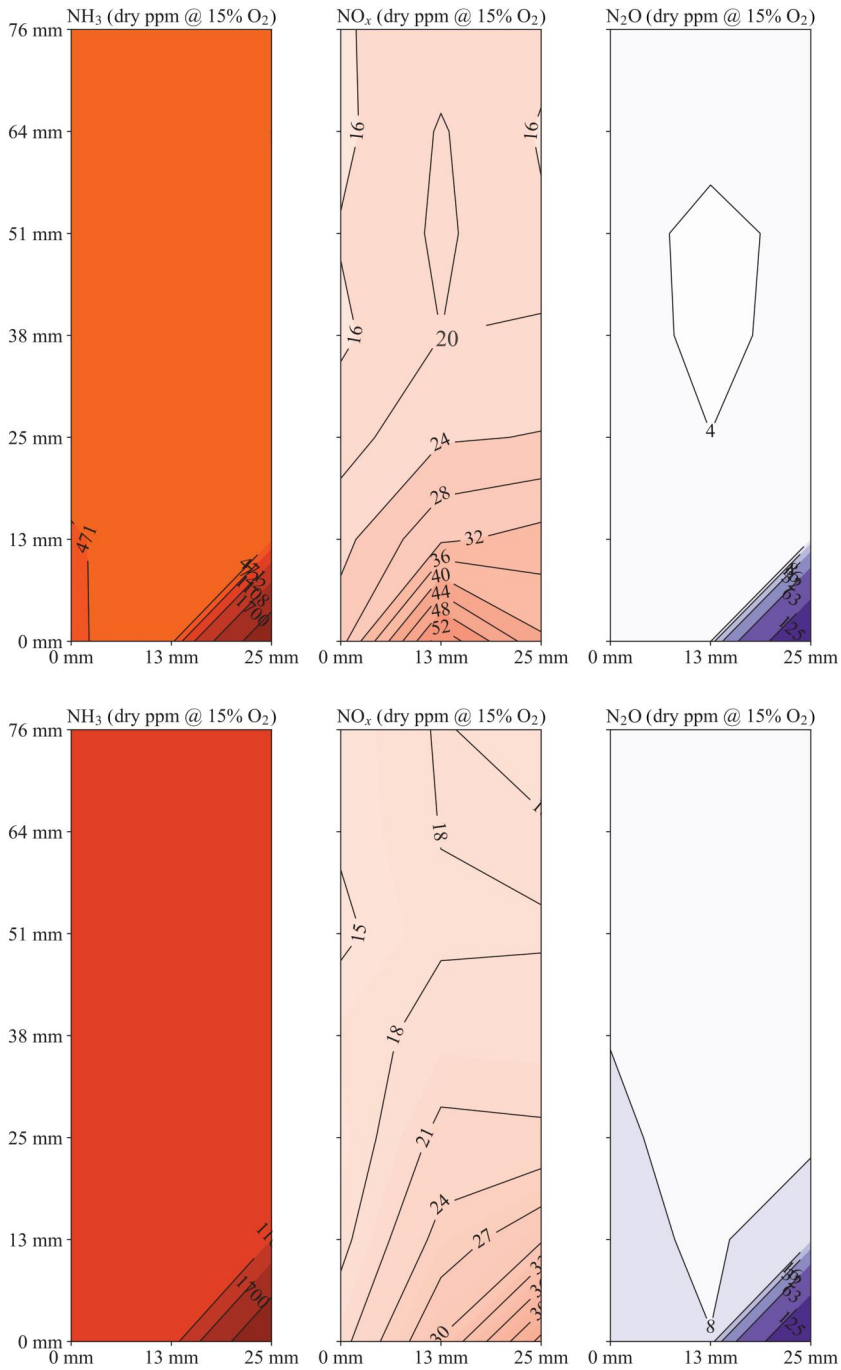

Fig. 8 NO_x, N_2O, and NH_3 emissions at 15% dry O_2 corrected for $S_g1.1$–*S16* (top) and $S_g1.1$–*C8* (bottom) at $\phi = 1.15$

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combustion efficiency. When compared with the isocontours in Fig. 5, the area in contact with the quartz liner is decreased for $S_g1.1$–*S16* compared to $S_g1.1$–*C8*. Furthermore, the general flame shape transitions from a characteristic convex shape to a concave shape with increased interaction on the quartz liner. NH_3 emissions are elevated with the $S_g1.1$–*C8* geometry, which are double that of the $S_g1.1$–*S16* geometry for the same equivalence ratio, despite starting at similar levels in the flame brush. The increased unburned fuel and combustion inefficiency may be correlated with the flame shape and heat losses due to contact with quartz. For the $S_g1.1$–*S16* geometry, the increased combustion efficiency observed from the NH_3 emissions suggests a higher concentration of H_2 available for the secondary stage of the RRQL system, and minimal NH_3 for a lower contribution of fuel–bound NO_x to the total staged combustion emissions.

In terms of N_2O, the emissions trends are slightly elevated at the quartz walls of the burner, particularly for the $S_g1.1$–*C8* geometry, indicating lower temperatures and higher wall heat losses. By comparison, the $S_g1.1$–*S16* swirler geometry has a more uniform distribution of N_2O across the radial point positions.

4 Conclusions

An experimental study was carried out to understand the impact of the swirl pattern on the flame morphology and exhaust emissions of rich premixed NH_3–air flames. Focus was on the characterization of the combustion zone through flame imaging and exhaust emissions. The main conclusions are:

(1) Swirler geometry can dramatically affect flame shape, peak radical concentrations, and emissions species for rich NH_3 flames.
(2) Elevated heat losses caused by swirler–liner interaction, indicated by regions of high OH* intensity, can cause the optimal ϕ to shift closer to stoichiometric due to elevated NH_3 and N_2O emissions at richer ϕ as shown by the $S_g1.1$–*C8* swirler configuration.
(3) This leaner ϕ for $S_g1.1$–*C8* suggests a potentially lower H_2 concentration for the second combustion stage, which could result in reduced overall combustion efficiency.
(4) The elevated N_2O concentrations are likely attributed to heat losses through the walls, potentially exacerbated by the higher flame to wall interaction area produced by the lobed flame structures.
(5) Reduced NH_3 cracking in the primary stage implies a higher contribution of fuel–bound nitrogen (fuel–NO_x) to overall NO_x production in the secondary lean stage.

These findings highlight the intricate interplay between swirler geometry, flame structure, and emission characteristics, underscoring the importance of careful optimization for minimizing emissions and maintaining combustion efficiency. As a result, we chose the $S_g1.1$–*S16* swirler for further RRQL studies (Part II of this paper).

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Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

CO = carbon monoxide
CH_4 = methane
FTIR = Fourier-transform infrared
H_2 = hydrogen
NH_3 = ammonia
NO_x = nitrogen oxide
N_2O = nitrous oxide
RRQL = rich–relaxation–quench–lean
S_g = geometric swirl number
U_0 = bulk velocity
ϕ = equivalence ratio

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